PYTHON PROGRAMMING | EXERCISE 02

Finite State Automaton Ending in ab

Recognizing strings over the alphabet a and b

The DFA tracks whether the input currently ends in no useful suffix, 'a', or the accepting suffix 'ab'. Any symbol outside the alphabet is rejected.

Question

Implement a basic finite state automaton that recognizes strings ending with 'ab' using Python.

Code

```
class EndsWithAB:
    transitions = {
        "q0": {"a": "q1", "b": "q0"},
        "q1": {"a": "q1", "b": "q2"},
        "q2": {"a": "q1", "b": "q0"},
    }
    accepting_state = "q2"

    def accepts(self, text):
        state = "q0"
        for symbol in text:
            if symbol not in {"a", "b"}:
                return False
            state = self.transitions[state][symbol]
        return state == self.accepting_state


dfa = EndsWithAB()
print("DFA accepts exactly strings ending in 'ab':")
for sample in ["ab", "aab", "babab", "aba", "abb", "ba", ""]:
    result = "ACCEPT" if dfa.accepts(sample) else "REJECT"
    print(f"{sample!r:8} -> {result}")
```

Output

```
DFA accepts exactly strings ending in 'ab':
'ab'     -> ACCEPT
'aab'    -> ACCEPT
'babab'  -> ACCEPT
'aba'    -> REJECT
'abb'    -> REJECT
'ba'     -> REJECT
''       -> REJECT
```

Output screenshot

